

Even Synthesis Is Transport: At 3B, Implicit Multi-Hop Reasoning Reduces to Retrieval-and-Elimination, and Composition Appears Only as Externalized Self-Delivery

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Abstract

Retrieval can be explained as transport: a prior program showed that long-context retrieval failure is repaired by *delivering* a fact, not by redirecting attention or by a reusable internal correction. Synthesis is where that account should break: in an $A \rightarrow B \rightarrow C \rightarrow D$ chain the target D is *absent* from context and must be *assembled*, so a model that only transports cannot answer — unless it composes. We test, pre-registered and cross-model (Llama-3.2-3B, Qwen2.5-3B; greedy; both vanilla bases), *when* D forms (prefill vs. decode) and *whether the stated hops are used*, with four locked controls and a built-in positive control. The verdict replicates on both models. **Implicit (no scratchpad — the clean computation test): reduces to delivery.** Three independent operationalizations agree: (i) in-place *corruption* of the middle hop flips the answer in only 5%/11% of cases the model gets right, while corrupting the code fact flips 100%; (ii) masking the link facts at the answer-emitting step is inert, masking the code kills it; (iii) blocking the elimination shortcut with decoys floors implicit accuracy to chance ($0.35 \rightarrow 0.07$, $0.45 \rightarrow 0.10$). The model answers by *retrieval-and-elimination* over competitor facts, not by traversing the stated edges. **Explicit chain-of-thought: genuine traversal — but scaffolded by self-delivery.** Writing the trace makes every hop causal and corruption disrupts the answer (84%/90% middle-hop flip — the positive control proving the implicit null is a real null, not a broken test), and it is decoy-robust; but the model composes by writing each intermediate fact *into* context and reading it back — converting computation into delivery (self-RAG), not internal decode-time binding. We also surface a methodological catch: under competition, node *removal* is confounded by elimination (a removed mid-fact is recoverable from competitors), so removal under-states hop usage; in-place *corruption* is the faithful test. Conclusion, regime-bounded: at 3B, internal multi-hop computation of an absent fact does not occur; “reasoning” here is retrieval-and-elimination, and real composition appears only as externalized self-delivery. This closes the synthesis regime in the same direction as attention (γ : transport) and the delivery ladder (repair = deliver the binding): in this model class the network *delivers and composes through an external buffer; it does not latently compute*. The null is the strongest of the three, because synthesis is the one place transport could have failed — and it did not. Code, pre-registration, and raw per-case logs are released.

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1 Introduction

Two prior results in this program localize the mechanism of long-context behaviour to *transport*. Up-weighting a needed key’s attention buys nothing beyond plain re-delivery, so attention is weighted transport, not computation [1]; and the repair for retrieval-disambiguation failure is delivering the attribute→code *binding* — not attention redirection and not a transplanted state correction, both of which are closed [2]. Each was a retrieval task: the answer is *present* in context, and the question is whether the model moves it to the output.

Synthesis is the adversarial case for that account. In a composed chain $A \rightarrow B \rightarrow C \rightarrow D$ where D is bound only to C and the question names only A , the answer D is *not present anywhere*; it can only be produced by chaining the hops. A pure transporter has nothing to transport. If the program’s “everything is transport” reading is going to fail, it should fail here. We therefore make synthesis the decisive test, and we sharpen the question from “does synthesis happen” to *when does the target form and are the stated hops used*:

- **P1 decode-computation:** the binding $A+B+C \rightarrow D$ happens at the answer step — the positive mechanism we would hope to find.
- **P2 prefill-assembly:** D is already encoded during reading; decode merely emits it — still transport, of a representation formed at read-time.
- **P3 reduces-to-delivery:** the “chain” is not traversed — a single code fact plus elimination over competitors solves it — retrieval in disguise.

We pre-register four controls to separate these (necessity of each node; a false-hop usage test; mask-at-answer for prefill-vs-decode; a decoy sweep that blocks elimination), run both ported 3B models, and report implicit and chain-of-thought separately because explicit CoT writes intermediate facts into context and thereby *is* a form of delivery. The result, replicated on both models, is P3 for implicit and self-delivered traversal for CoT.

Relation to the program. This is the capstone of a three-gate arc: γ (attention = transport, [1]), the delivery ladder (repair = delivery, [2]), and synthesis (here). The first two concern retrieval; this paper contributes the synthesis gate and the cross-arc thesis. The unifying claim — *this model class delivers and composes via an external buffer but does not latently compute* — now rests on three independent causal gates, cross-model.

2 Related work

Multi-hop reasoning and latent composition. Whether transformers compose facts internally or rely on shortcuts is actively debated; work on latent multi-hop reasoning finds it weak and entity-dependent [3], and the “grokked transformers” line shows implicit composition emerging only under heavy training [4]. Our contribution is a *causal* usage test (corruption, mask-at-answer) on off-the-shelf 3B models, with a decoy control that isolates an elimination shortcut. **Chain-of-thought as externalization.** CoT improves multi-step accuracy [5]; analyses question whether the written trace is faithful to the computation [6]. We give a mechanistic reading: in this regime CoT *is* the mechanism — the model composes by writing and re-reading, self-delivery rather than internal binding. **Shortcuts and elimination.** Models exploit answer-set structure and process-of-elimination [7]; we show the implicit “chain” success is largely this, and that it collapses when decoys remove the eliminable structure. Methodologically we follow a matched-magnitude control discipline [8] and build on the program’s retrieval results [1, 2].

3 Design

Chain. Synthetic $A \rightarrow B \rightarrow C \rightarrow D$ (attribute → person → zone → 7-digit code), D bound *only* to C ; the question gives A , so only the composed chain reaches D . $K=3$ (2 distractor

Table 1: Synthesis gates, $n=60$. “causal” = removal-drop CI excludes 0. corr-flip = $P(\text{answer leaves } D \mid \text{baseline-correct})$ under in-place hop corruption. Recomputed from the per-cell logs.

cell	baseline	Gate0 AB/BC/CD	flip AB/BC/CD	follow-false BC	decoy 0→10
Llama implicit	0.350	F / F / T	0.19 / 0.05 / 1.00	0.00	0.35 → 0.07 ↓
Qwen implicit	0.450	T / F / T	0.07 / 0.11 / 1.00	0.04	0.45 → 0.10 ↓
Llama CoT	0.617	T / T / T	0.73 / 0.84 / 1.00	0.81	0.62 → 0.58 ≈
Qwen CoT	0.650	T / T / T	0.87 / 0.90 / 1.00	0.79	0.65 → 0.72 ≈

chains for competition), hops scattered among filler at $N=4096$, SEED=20260531, $n=60$ per cell, paired bootstrap (10^4) 95% CIs. Both vanilla base models (bf16, sdpa, greedy); no sparse kernel, no patch, no sealed-component edit — the object is the *model’s* synthesis of an absent fact. Gate-2 masking is validated (empty-mask $\equiv$ full, bit-identical). A pilot certified all four (model $\times$ mode) cells informative (dense accuracy in $[0.30, 0.95]$).

Two modes, never merged. *Implicit* (no scratchpad) is the clean internal-computation test. *Explicit CoT* is the contrast — but, as pre-stated, writing the trace puts intermediate facts *into* context, so CoT measures “can the model use its own scratchpad”, a delivery-flavoured competence.

Four pre-registered controls. (0) *Necessity*: remove each node; a node is causal iff removal drops dense accuracy (CI excludes 0). (1) *Usage / false-hop*: corrupt a hop in place to a false trail; a traverser must follow it. (2) *Formation*: mask a node at the answer-emitting step — if the answer survives, D formed at prefill (P2); if it dies at the causal node, binding needs the node at decode (P1). (3) *Decoy sweep*: add code-bearing decoy zones that block the elimination shortcut. The verdict map (P1/P2/P3) was fixed before any result.

4 Results

Both models replicate (Table 1). We report implicit (the clean test) first, then CoT (which doubles as the positive control).

4.1 Implicit reduces to delivery (P3), three ways

(1) Corruption: the links are not used. On cases the model answers *correctly*, corrupting the middle hop ($B \rightarrow C$, the zone) to a false trail flips the answer in only 5% (Llama) / 11% (Qwen) of cases, and the model follows the false trail 0%/4% of the time. Corrupting the code fact ($C \rightarrow D$) flips 100% (follow-false 81%/85%). Figure 1: the model reads the code but does not use the link facts, even when it is right. By the pre-registered rule (still-correct under a false mid-fact $\Rightarrow$ the hop was not used $\Rightarrow$ not traversed) this is P3.

(2) Mask-at-answer: only the code is consulted. Masking $A \rightarrow B$ or $B \rightarrow C$ at the answer-emitting step is inert (accuracy $\approx$ full $\approx$ matched-filler control); masking $C \rightarrow D$ drives accuracy to 0.000 (Fig. 3). The answer step consults only the code fact: there is no decode-time binding of the absent D to disrupt.

(3) Decoy sweep: the elimination crutch is load-bearing. Removing the target’s mid-hop still leaves the *competitors’* facts, so the model can deduce the target zone by elimination (“not in the other persons’ zones”). Adding decoy zones (a code fact with no person) makes the target indistinguishable once its mid-hop is gone, blocking elimination. Under decoys, implicit accuracy floors to chance (Llama 0.35→0.07, Qwen 0.45→0.10), while CoT stays flat (Fig. 2).

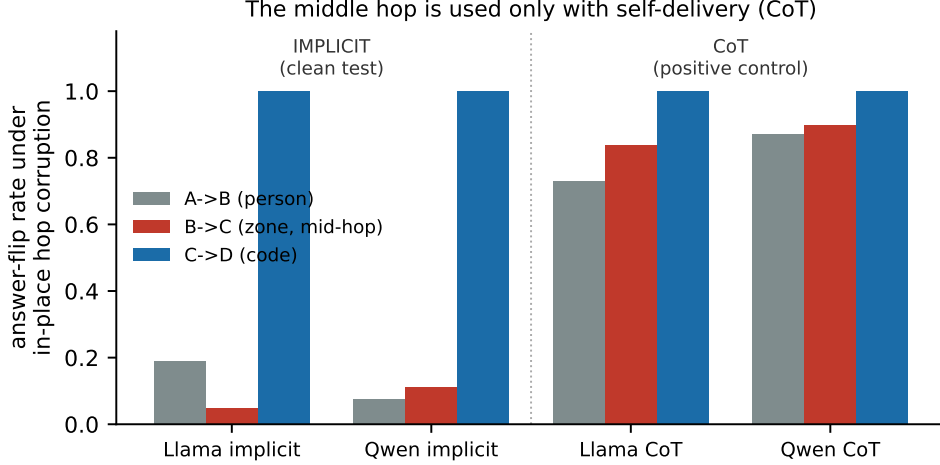


Figure 1: The decisive panel. Corrupting the code fact ($C \rightarrow D$, blue) flips the answer everywhere — the code is read. But corrupting the middle hop ($B \rightarrow C$, red) flips the implicit answer essentially never (0.05/0.11) while flipping the CoT answer 0.84/0.90. The same corruption machinery detects hop usage under CoT, so the implicit null is a real null. The hop is used only when the model writes it down.

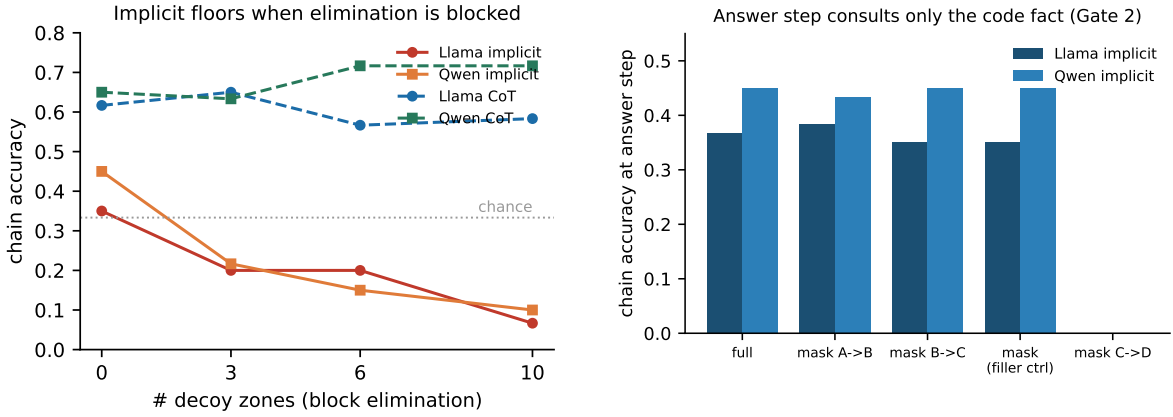


Figure 2: Decoy sweep. Blocking elimination floors implicit to chance (dotted) while CoT is flat — an orthogonal confirmation with its own control.

Figure 3: Gate 2 (implicit). Masking the link facts at the answer step is inert; only masking the code fact kills the answer. No decode-time chain binding to disrupt.

The flat CoT sweep is the critical control: identical decoys do not hurt a model that actually traverses, so the implicit collapse is specifically the loss of retrieval-and-elimination, not added distraction. Implicit success $\subset$ elimination.

4.2 Explicit CoT: genuine traversal, scaffolded by self-delivery

With a written trace all three hops become causal (removal drop +0.52/+0.57 for the mid-hop), corruption disrupts every hop, and the model follows a false mid-trail 81%/80% of the time — it genuinely routes through “ B is in C ”. It is decoy-robust. But this is the pre-registered caveat: the model emits B , then C , then D as tokens, each delivered back into context and re-read. It composes by *externalizing* the hops (self-RAG), not by an internal decode-time binding. Mask-at-answer is undefined for CoT (the answer token is many steps after the written hops), so CoT sits outside the P1/P2/P3 trichotomy specific to the clean implicit test and is reported as traversal-via-self-delivery. Its role here is twofold: a demonstration that the corruption machinery *can* detect traversal (the positive control for §4.1), and a mechanistic

statement that the only way these models traverse the chain is to write it down.

4.3 A methodological catch: removal is confounded by elimination

The pre-registration’s literal Gate 0 (node *removal*) under-states hop usage under competition: a removed mid-fact is recoverable from the competitors, so removal can read “non-causal” even for a used hop. The faithful test is in-place *corruption* (a false trail a traverser must follow), which cannot be elimination-recovered. The removal $\leftrightarrow$ corruption dissociation is itself informative: Qwen-implicit’s $A \rightarrow B$ is “causal” by removal (-0.18 , it removes the query anchor) yet inert under corruption (flip 0.07) — the model uses the A -fact’s *presence* as a structural anchor, not the named entity. The critical evidence mass for implicit success is essentially $\{C \rightarrow D\}$ alone.

5 Discussion

The arc closes on one locus, three times. Attention redirection is transport [1]; stored-state correction is not reusable and the repair is delivery of the binding [2]; and now synthesis of an absent fact is, internally, retrieval-and-elimination, with real composition appearing only when the steps are externalized. Three independent causal gates, cross-model over two families and 1B–3B, converge: *this model class delivers, and composes through an external buffer; it does not latently compute*. The synthesis null is the strongest leg because synthesis is the one regime where a pure transporter *should* have failed — the target is absent — and the transport reading still held.

Why this is a solid null, not a floor artifact. The implicit null is conditioned on baseline-*correct* cases: even when the model gets the chain right, it did so without using the middle hop. The same corruption test detects usage under CoT (0.84 – 0.90), so the implicit 0.05 – 0.11 is a real null and not a broken probe. The decoy sweep adds an orthogonal manipulation with its own flat-CoT control. Three operationalizations, a working positive control, and an orthogonal manipulation all agree. As the pre-registration demanded, a positive after a null streak warrants extra scrutiny — and the one place a positive appeared (CoT) is correctly attributed to self-delivery, not internal computation.

Product reading. Consistent with the program: for these model sizes do not expect latent multi-step reasoning over delivered facts. Get composition by *externalizing* it — prompt the model to write its steps (CoT), or run an explicit retrieve-compose-retrieve loop (self-RAG / scratchpad) — rather than hoping for an internal chain. “Smarter” on a fixed model comes from the scaffold or from capacity, not from an internal shortcut.

6 Limitations

- **Regime.** Synthesis, **3B**, synthetic attr $\rightarrow$ person $\rightarrow$ zone $\rightarrow$ code chains, $N=4096$, $K=3$, greedy. A null is “no decode-computation at 3B on this design”, *not* “never”. Larger models, other chain types, and trained composition are separate gates; the null does not transfer up.
- **Low implicit baselines.** Implicit accuracy ($0.35/0.45$) is only modestly above chance (0.333); the clean test sits near the floor edge. The verdict is corruption-based and baseline-correct-conditioned, robust to the low baseline, but a higher-headroom implicit chain would strengthen it.
- **CoT is not internal computation.** CoT “traversal” is via self-delivery (writing the steps); Gate 2 is undefined for it, so it is evidence of externalized composition, not of decode-time binding.

- **Open channel.** How implicit binds *A* to the right code *above chance* without the named hops (some prefill-formed, elimination-assisted, structural association) is left open — it is not a traversal of the stated edges, but its exact form is uncharacterized.
- **Off-distribution interventions.** Mask-at-answer and corruption are off-distribution; degeneration is expected and was required to be distinguished from clean answer-death (specific wrong-endpoint), not miscounted.

7 Conclusion

At 3B, implicit multi-hop synthesis of an absent fact does not occur: the model retrieves a code and resolves the competition by elimination; corrupting the chain’s links leaves correct answers untouched, masking them at the answer step is inert, and blocking elimination floors accuracy to chance. The model traverses the chain only when permitted to *write the steps*, converting computation into delivery. Across attention, retrieval repair, and now synthesis — three independent causal gates, cross-model — the finding is one: this model class transports and composes through an external buffer; it does not latently compute. Internal decode-time composition, if it exists, lives above 3B or inside the scratchpad — a separate gate.

Reproducibility

The repository provides the pre-registration, the pilot/calibration and gate harnesses, the raw per-case logs and per-cell gate aggregates for all four (model $\times$ mode) cells, the chain-results stream, and the masking sanity gate; `make_figures.py` recomputes every figure from the gate aggregates. Llama-3.2-3B and Qwen2.5-3B; single RTX 4060 Ti.

References

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